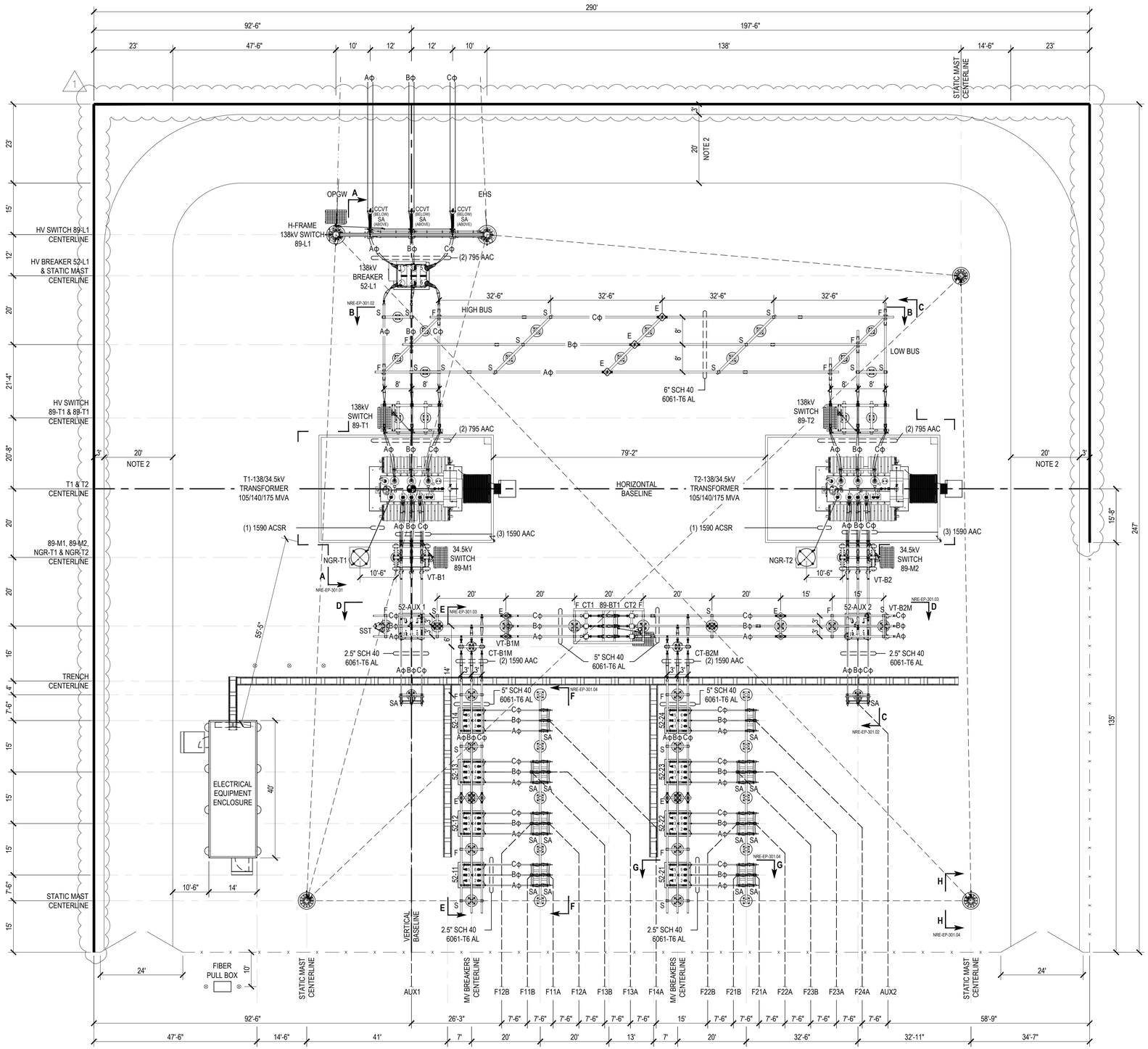

GENERAL ARRANGEMENT

SCALE: 1/16" = 1'-0"

NORTH

NOTE:

1. SUBSTATION CLEARANCES AND SPACING LISTED BELOW, BASED ON IEEE STD 1427-2020 AND IEEE STD C37.30.1-2011 TABLE 14.

RATED VOLTAGE	34.5kV	138kV
RIGID BUS CENTER-TO-CENTER SPACING:	3'-0"	8'-0"
MINIMUM PHASE-TO-PHASE, METAL-TO-METAL DISTANCE:	1'-6"	5'-3"
MINIMUM PHASE-TO-GROUND CLEARANCE:	1'-3"	4'-2"
MINIMUM VERTICAL CLEARANCE TO LIVE PARTS:	10'-0"	12'-0"
MINIMUM BOTTOM OF INSULATOR OR BUSHING PORCELAIN TO TOP OF FOUNDATION:	8'-6"	8'-6"
MINIMUM HORIZONTAL CLEARANCE TO FENCE FROM LIVE PARTS:	10'-7"	14'-0"
MINIMUM VERTICAL CLEARANCE TO FENCE:	15'-0"	17'-1"
MINIMUM OVERHEAD CONDUCTOR TO ROADWAYS INSIDE SUBSTATION	22'-0"	25'-0"

BUS	HEIGHT
MV MAIN BUS (5" SCH 40 6061-T6)	27'-0"
MV FEEDER BUS (2.5" SCH 40 6061-T6)	21'-0"
MV TRANSFORMER	24'-0"
HV HIGH BUS	25'-0"
HV LOW BUS	15'-0"

2. DRIVE PATHS SHOWN FOR ILLUSTRATION PURPOSES ONLY.
3. HIGH TEMPERATURE COMPOUND TO BE USED AT ALL PAD CONNECTIONS.
4. EQUIPMENT LAYOUT COULD CHANGE BASED ON VENDOR DRAWINGS.
5. HV LINE SWITCHES, ARRESTERS, AND CCVTS ARE MOUNTED ON THE H-FRAME AND ARE ALIGNED WITH THE INCOMING T-LINE.
6. FUTURE STUDIES MAY REQUIRE ADDITIONAL EQUIPMENT WITHIN THE SUBSTATION.

BUS AND BUS DAMPERS:

1. 1/4" WEEP HOLES SHALL BE PLACED IN THE CENTER OF EACH BUS RUN (BETWEEN 2 BUS SUPPORTS). THE WEEP HOLE SHALL BE AT THE BOTTOM OF THE BUS.
2. BUS DAMPERS WILL BE REQUIRED AND INSTALLED INSIDE TUBE BUS FOR ANY LENGTH EXCEEDING WHAT IS SHOWN ON THE TABLE BELOW (UNLESS STATED OTHERWISE):

TUBULAR BUS SIZE	MAX SPAN LENGTH BEFORE A BUS DAMPER IS REQUIRED (WITHOUT SUPPORTS)	BUS DAMPER SIZE
2.5"	10'-9"	477 AAC
5"	21'-3"	795 AAC

3. BUS DAMPER SHALL BE CLAMPED ON BOTH ENDS TO PREVENT BIRDCAGING.

LEGEND:

- F — FIXED FITTING
- S — SLIP FITTING
- E — EXPANSION FITTING
- SA — SURGE ARRESTER
- PEDESTRIAN RATED CABLE TRENCH SECTION
- FENCE
- 3/8" EHS
- OPGW
- CONCRETE WALL
- FEEDER CONDUIT
- LED LIGHT FIXTURE
- GROUNDING MAT
- BOLLARD
- ORIGIN POINT

REV	DATE	DESCRIPTION	DSGN	APPD
1	03/09/2026	GROUNDING UPDATES	CAS	CBH
0	01/28/2026	ISSUED FOR CONSTRUCTION	CAS	CBH

ENGINEER'S STAMP

Mortenson
700 MEADOW LANE NORTH
MINNEAPOLIS, MN 55422

spearmint energy

DATE: 01/28/2026
DRAWN BY: SOA
DESIGNED BY: CAS
REVIEWED BY: JCJ

PROJECT:
NOMADIC RED EGRET
TEXAS CITY, TEXAS
138/34.5kV SUBSTATION

DWG NAME:
138/34.5kV SUBSTATION
GENERAL ARRANGEMENT

MORTENSON ENGINEERING SERVICES, INC. TEXAS BOARD OF PROFESSIONAL ENGINEERS AND LAND SURVEYORS CERTIFICATE OF REGISTRATION #F-18964

INTERNAL PROJECT #: 23252035

DRAWING #: NRE-EP-300.01

REV 1

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